

Generative Proposals (Candidate Pool N_0)

Stage 1: Syntactic & AST Parsing

Checks: Syntax valid, elaborated module hierarchy, port types
Engines: Tree-Sitter / Verilator / Slang • Cost: Milliseconds (free / open-source)

Stage 2: Interface Schema & Interconnect

Checks: AXI / TLM handshakes, port widths, clock domain tags
Engines: Static Schema Linters / Protocol Checkers • Cost: Seconds (low compute overhead)

Stage 3: Functional & Assertion Verification

Checks: Temporal invariants, state-machine deadlocks, coverage
Engines: SystemVerilog Assertions (SVA) / BMC / Sim • Cost: Minutes (simulation / SAT)

Stage 4: Static Timing Analysis (STA)

Checks: Multi-corner setup & hold slack, WNS/TNS
Engines: OpenSTA / PrimeTime / Tempus • Cost: Minutes to hours (licensed)

Stage 5: Physical DRC / LVS Signoff

Checks: Design rules (DRC), layout vs. schematic (LVS)
Engines: Calibre / KLayout / Netgen • Cost: Hours+

Signoff-Clean Implementation (GDSII / OASIS)

Near-Miss Timing & DRC Failures
Route to Localized ECO Optimization
(Buffer insertion, gate sizing, wire widening)